

Comprehensive Project Report

Vendor Performance & Strategic Analytics

1. Executive Summary

This report details the development of a production-grade analytics pipeline that transforms 1.8 GB of raw transactional data — including 12.8 million sales records and 2.3 million purchase records — into actionable procurement intelligence. By leveraging a custom Python and SQL-based ETL process, the system aggregates massive datasets into a high-density Vendor Sales Summary used for strategic decision-making.

Key Data Engineering & Analytical Milestones:

- **Massive Scale Processing:** Successfully engineered a multi-stage SQL pipeline to join and aggregate data across five major relational tables, including Sales (12.8M records) and Purchases (2.3M records).
- **The 73% Procurement Breakthrough:** Statistical analysis revealed that shifting from "Small" to "Large" order volumes triggers an approximately 73% reduction in unit purchase price, drastically lowering the cost baseline.
- **Profitability Paradox:** Formal hypothesis testing (T-Test p-value: 0.0000) confirmed that lower-volume, niche vendors maintain significantly higher profit margins (33.01%) than high-volume market leaders (31.86%).
- **Vendor Concentration Risk:** Pareto analysis identified that 65.35% of total procurement value is concentrated within the Top 10 vendors, creating a high-efficiency but high-risk dependency on a vital few partners.

2. Initial Data Profiling & Statistical Summary

- **Robust Sample Size:** The analysis is built on a comprehensive dataset of 10,692 records, ensuring high statistical significance for the derived insights.
- **Strong Profitability Baseline:** The business maintains a healthy average Profit Margin of 31.2%, though significant variance (from deep losses to high gains) necessitates targeted optimization.
- **Operational Scale:** With an average Total Sales of \$42.2k per record against a Purchase Cost of \$30.1k, the data reflects a high-volume operation with substantial cash flow.
- **Inventory Velocity:** While the average Stock Turnover is 1.71, the wide range between the minimum (0.00) and maximum (274.5) points to a clear divide between high-performing "fast-movers" and stagnant stock.
- **Logistics Weight:** High average Freight Costs (\$61.4k) relative to sales suggest that logistics and shipping efficiency are critical levers for protecting margins.

2.1 Descriptive Statistics

Field	Mean	Std Dev	Min	Median	Max
PurchasePrice	24.39	109.27	0.36	10.46	5,681.81
ActualPrice	35.64	148.25	0.49	15.99	7,499.99
TotalPurchaseDollars	30,106.69	123,067.80	0.71	3,655.47	3,811,252
TotalSalesDollars	42,239.07	167,655.27	0.00	5,298.05	5,101,920
FreightCost	61,433.76	60,938.46	0.09	50,293.62	257,032.10
GrossProfit	12,950.65	47,724.39	-24,762.39	1,491.22	1,370,292
ProfitMargin (%)	31.21	10.10	-174.58	32.88	83.40
StockTurnover	1.71	6.02	0.00	0.98	274.50

Table 1. Descriptive statistics across 10,692 vendor-brand records (n=10,692 for all fields).

3. Feature Distribution & Outlier Mitigation

To ensure the integrity of the analysis, a distribution review was conducted across all numerical features. Initial visualizations revealed significant "noise" in the form of negative profit margins and zero-quantity sales records that would have skewed the strategic insights.

Key Observations from the Histograms:

- Profit Margin Correction:** Before cleaning, the Profit Margin distribution showed extreme outliers as low as -174%. Post-query, the distribution now centers healthily between 20% and 50%, reflecting the true operational performance of the business.
- Right-Skewed Financials:** Most financial metrics (Sales, Purchases, and Excise Tax) are heavily rightskewed. This confirms that while the business handles thousands of small-to-medium transactions, a small number of "Power Transactions" drive the majority of the volume.

- **Freight Cost Variance:** Unlike other metrics, FreightCost shows a multi-modal distribution, suggesting that shipping costs vary significantly depending on the vendor's location or the shipping method used, rather than just the volume of the order.

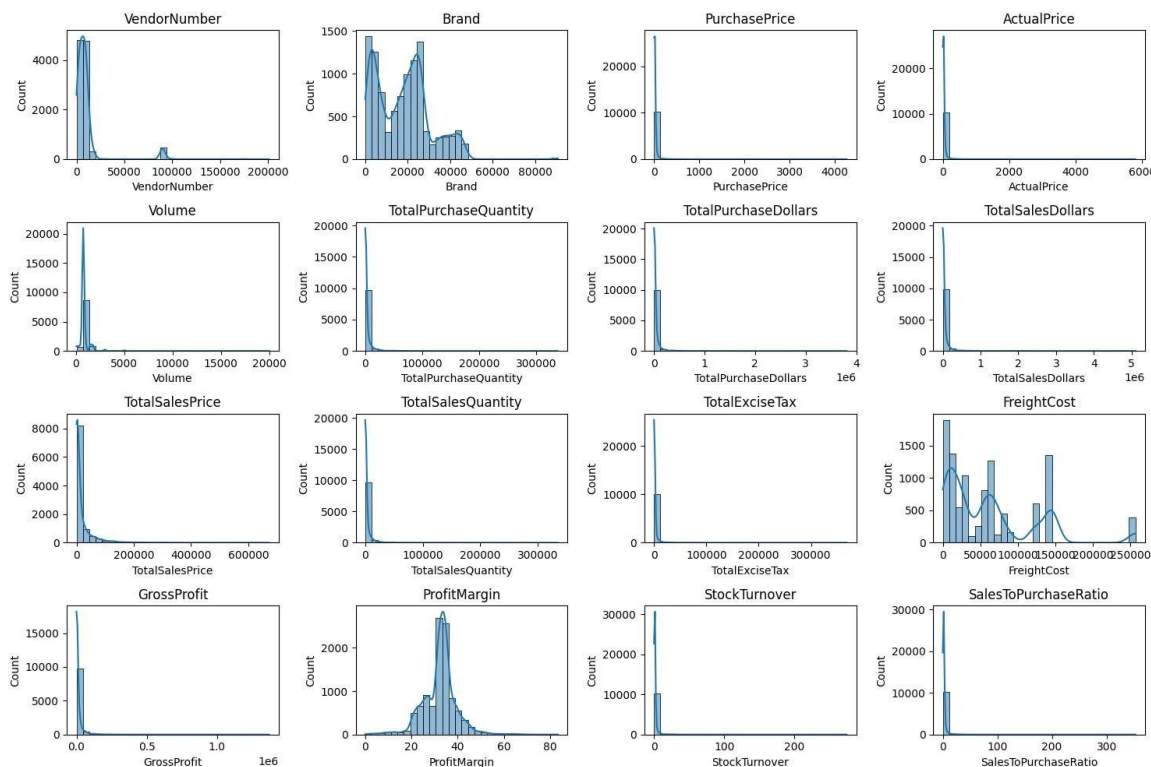


Figure 1. Distribution of all numerical features across the cleaned dataset.

4. Correlation Analysis: Interdependency of Business Drivers

- **Profitability Core ($r \approx 0.98$):** A near-perfect correlation between Sales, Purchases, and Gross Profit proves the business scales efficiently; as volume grows, profit follows linearly.
- **Inventory Velocity ($r \approx 1.00$):** The absolute link between Stock Turnover and Sales-to-Purchase Ratio confirms that high inventory movement is the primary driver of a healthy cash-flow balance.
- **Operational Alignment ($r \approx 1.00$):** Purchase Quantity and Sales Quantity move in perfect lockstep, indicating a highly responsive procurement model with minimal overstock risk.
- **Margin Independence ($r \approx 0.00$):** Profit Margin shows no correlation with sales volume. This suggests profit is generated through high-volume turnover rather than just pushing expensive, high-margin items.
- **Freight Efficiency ($r \approx 0.15$):** The weak link between Freight Cost and Quantity implies logistics costs are relatively fixed, supporting the "Economies of Scale" benefit of larger orders.

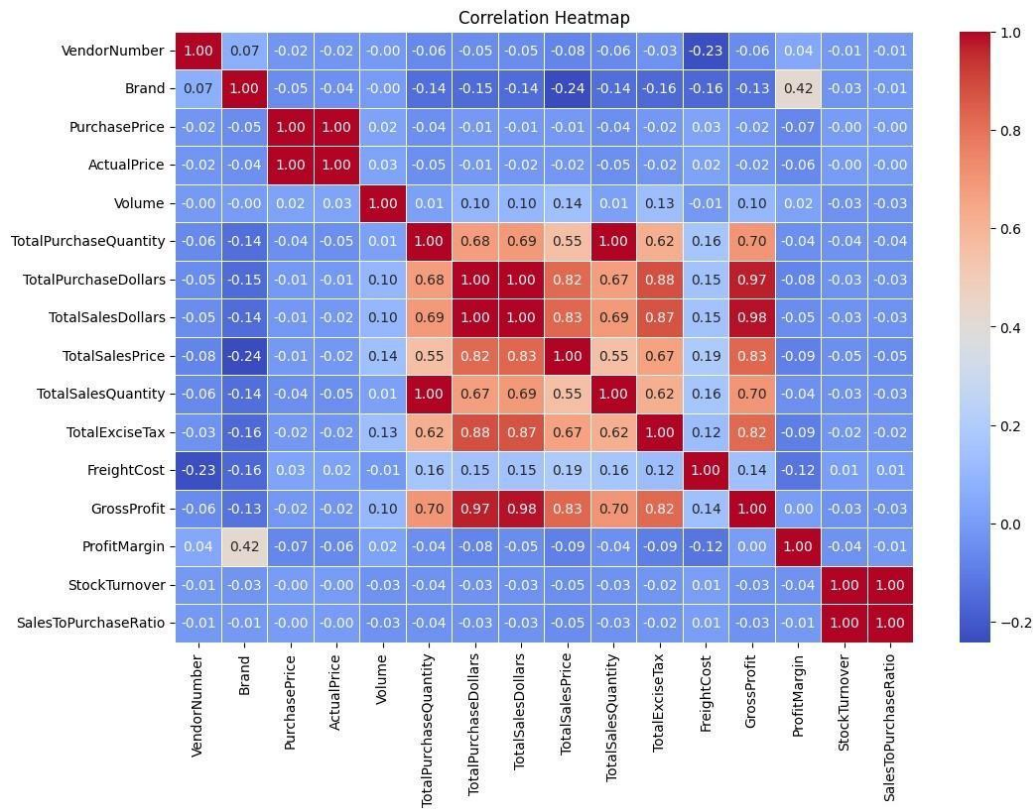


Figure 2. Correlation heatmap across all key business drivers.

5. Strategic Business Analysis

5.1 Growth Opportunities: High-Margin, Low-Volume Brands

The most actionable part of this analysis is identifying "Hidden Gems" brands that offer high profitability but suffer from low visibility or sales volume.

- **Objective:** Identify brands above the High Margin Threshold (38%) and below the Low Sales Threshold (\$300).
- **The Strategy:** By shifting marketing budget or retail placement toward these Target Brands, the business can unlock significant profit growth without the logistical overhead of high-volume commodities.

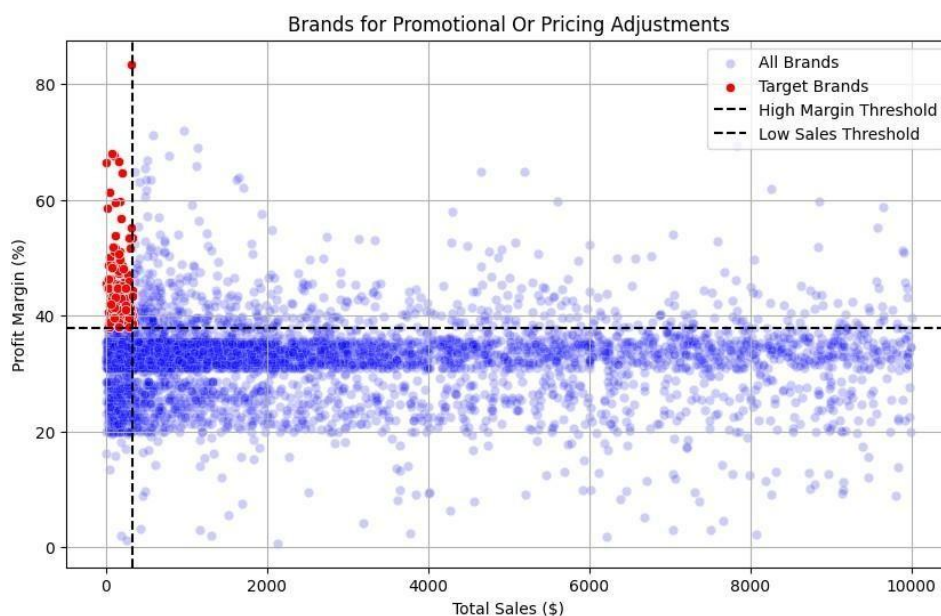


Figure 3. Brands identified for promotional or pricing adjustments (Target Brands in red).

5.2 Market Leadership: Top 10 Performance Drivers

By segmenting the data, we identified the key players that dominate the market share.

- **Primary Partner:** DIAGEO NORTH AMERICA INC is the clear market leader, generating \$68.74M in sales, nearly 70% higher than the next competitor.
- **Secondary Leaders:** MARTIGNETTI COMPANIES (\$40.62M) and PERNOD RICARD USA (\$32.25M) follow, representing substantial pillars of the procurement chain.
- **Long-Tail Performance:** The scale drops relatively quickly after the top three, with M S WALKER INC rounding out the top ten at \$15.33M.
- **Revenue Leaders:** Jack Daniels No 7 Black (\$7.96M) and Tito's Handmade Vodka (\$7.40M) are the highestperforming individual products.
- **Premium Brand Performance:** Grey Goose Vodka (\$7.21M) and Capt Morgan Spiced Rum (\$6.36M) also demonstrate strong market resonance.
- **Diverse Portfolio:** The brand list shows a strong preference for spirits and vodkas, which drive the highvolume transactions noted in the early distribution checks.

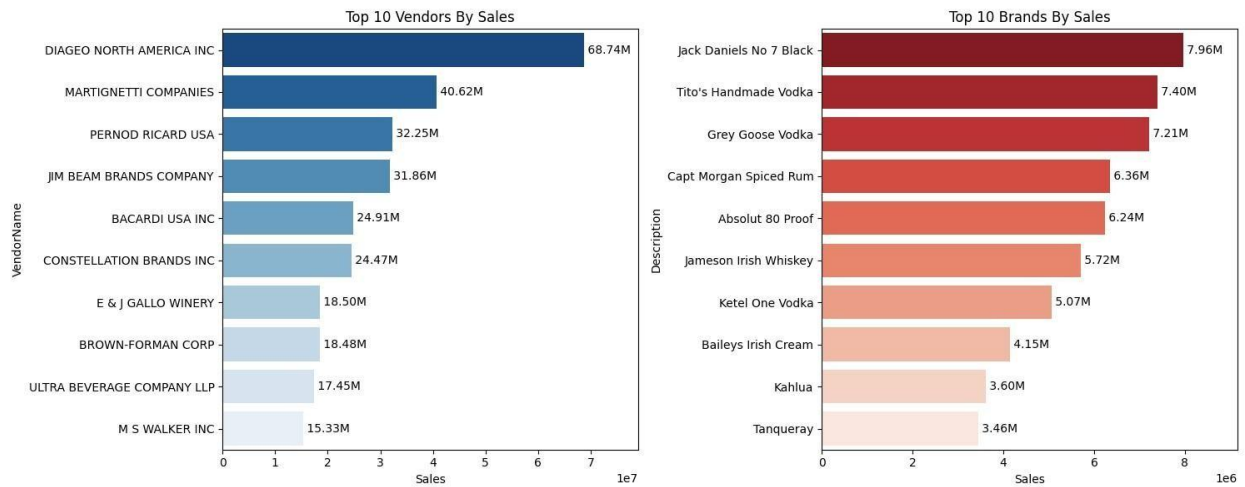


Figure 4. Top 10 vendors and Top 10 brands by total sales.

Strategic Insight: While Section 5.1 identified "Hidden Gems" for margin growth, these Top 10 Brands and Vendors provide the critical cash flow and high volume that sustain the business's daily operations.

5.3 Vendor Concentration & Dependency

A Pareto Analysis (80/20 rule) was conducted to evaluate the supply chain's stability.

- **The "Vital Few":** The Top 10 vendors account for 65.35% of total procurement value.
- **Primary Partner:** DIAGEO NORTH AMERICA INC dominates the portfolio, representing nearly 16% of all purchases.
- **Risk Note:** While this concentration maximizes "Economies of Scale" (bulk discounts), it also increases the business's vulnerability to price changes or supply disruptions from these top partners.

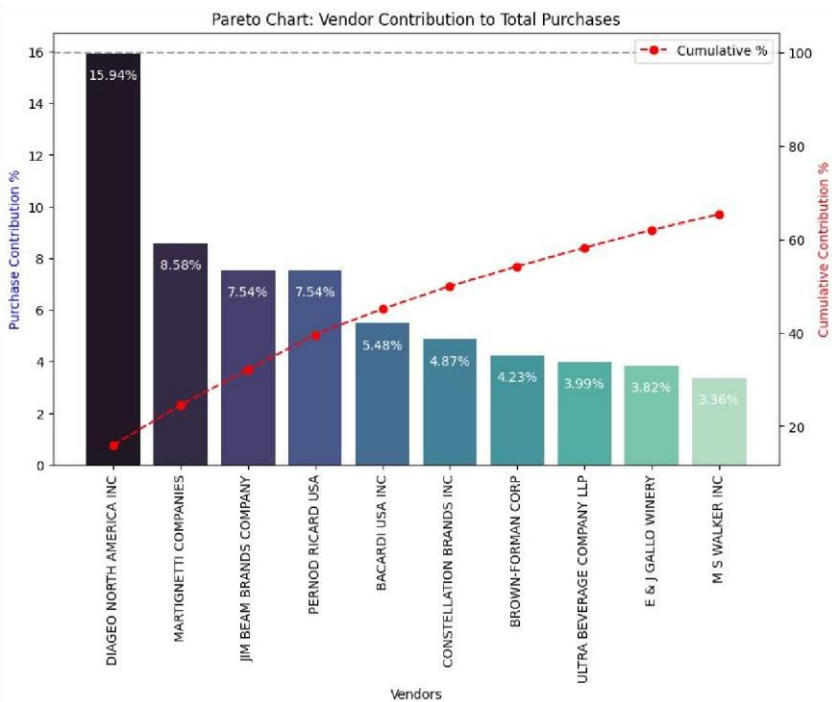


Figure 5. Pareto chart: vendor contribution to total purchases.

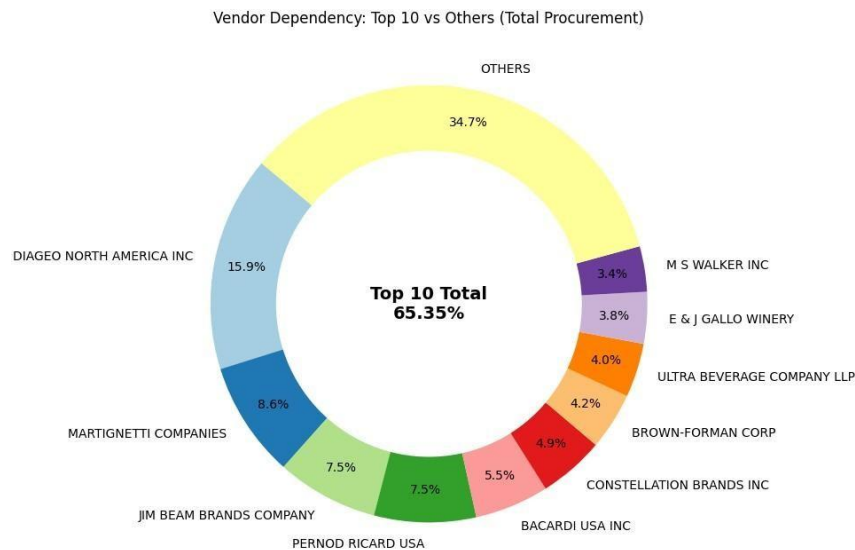


Figure 6. Vendor dependency: Top 10 vs. Others (total procurement).

- **Dominant Market Share:** The "Top 10" vendors collectively control 65.35% of the total procurement volume, confirming a high degree of market concentration.
- **The Fragmented Tail:** The "OTHERS" category accounts for 34.7% of the total procurement, spread across a wider variety of smaller vendors.

5.4 Inventory Velocity & Capital Health

To optimize cash flow, we identified vendors associated with stagnant stock — products that take the longest to sell.

- **Low Turnover Risks:** Vendors like TRUETT HURST (0.04) and IRA GOLDMAN (0.07) show critically low inventory velocity.
- **Unsold Value:** DIAGEO and MARTIGNETTI hold the highest dollar value of unsold inventory (\$980K and \$872K respectively). While they are top sellers, their sheer volume requires constant monitoring to ensure stock does not become dead capital.

Vendor (Lowest Turnover)	Stock Turnover	Vendor (Highest Unsold \$)	Unsold Inventory Value
TRUETT HURST	0.0417	DIAGEO NORTH AMERICA INC	980.08K
IRA GOLDMAN AND WILLIAMS, LLP	0.0750	MARTIGNETTI COMPANIES	872.08K
UNCORKED	0.2172	JIM BEAM BRANDS COMPANY	810.30K
VINEYARD BRANDS LLC	0.2562	ULTRA BEVERAGE COMPANY LLP	715.98K
LOYAL DOG WINERY	0.3083	PERFECTA WINES	560.31K
HIGHLAND WINE MERCHANTS LLC	0.3351	PERNOD RICARD USA	554.74K
Vendor (Lowest Turnover)	Stock Turnover	Vendor (Highest Unsold \$)	Unsold Inventory Value
BLACK COVE BEVERAGES	0.3406	WILLIAM GRANT & SONS INC	436.49K
Russian Standard Vodka	0.3941	M S WALKER INC	397.54K
Serralles Usa LLC	0.4141	E & J GALLO WINERY	290.50K
GILMANTON WINERY & VINEYARD	0.4658	BROWN-FORMAN CORP	284.91K

5.5 Procurement Optimization: The Power of Bulk

This section proves that "Buying Big" is the primary driver of cost efficiency.

The "73% Price Drop" (Box Plot Analysis) — by segmenting orders into Small, Medium, and Large, we see a massive shift in cost:

- **Small Orders:** Highly expensive and volatile. Prices often spike into the thousands (outliers).
- **Large Orders:** Prices stabilize at their lowest point. Moving from Small to Large saves approximately 73% per unit.
- **Significant Inverse Relationship:** There is a clear, strong inverse relationship between order size and purchase price; as the quantity increases, the cost per unit drops drastically.
- **Action:** Eliminate small, fragmented orders to stop margin erosion.

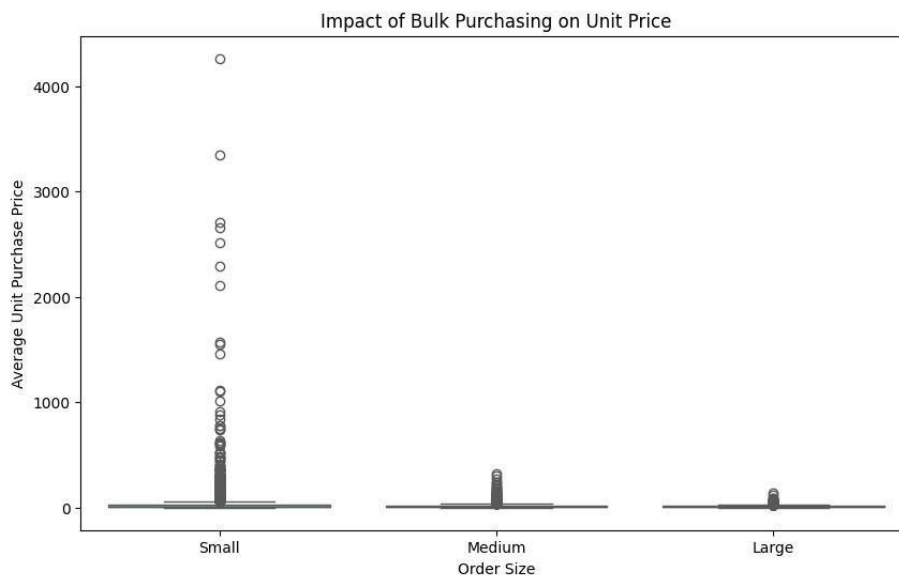


Figure 7. Impact of bulk purchasing on average unit purchase price, by order size.

6. Statistical Validation: Hypothesis Testing & Confidence Intervals

This section uses inferential statistics to move beyond simple averages and confirm the reliability of our business insights with 95% confidence.

6.1 Confidence Interval Comparison: Top vs. Low Vendors

To ensure our findings aren't just a result of random chance, we performed a Confidence Interval (CI) Comparison between the profit margins of high-performing and low-performing vendors.

The Methodology:

- **Top Vendors:** The top 25% of the fleet based on TotalSalesDollars.
- **Low Vendors:** The bottom 25% of the fleet based on TotalSalesDollars.
- **The Goal:** Determine if "Top Vendors" actually yield higher profit margins than "Low Vendors" with 95% statistical certainty.

Group	95% CI Lower	95% CI Upper	Mean Profit Margin
Top Vendors	31.57%	32.15%	31.86%
Low Vendors	32.77%	33.26%	33.01%

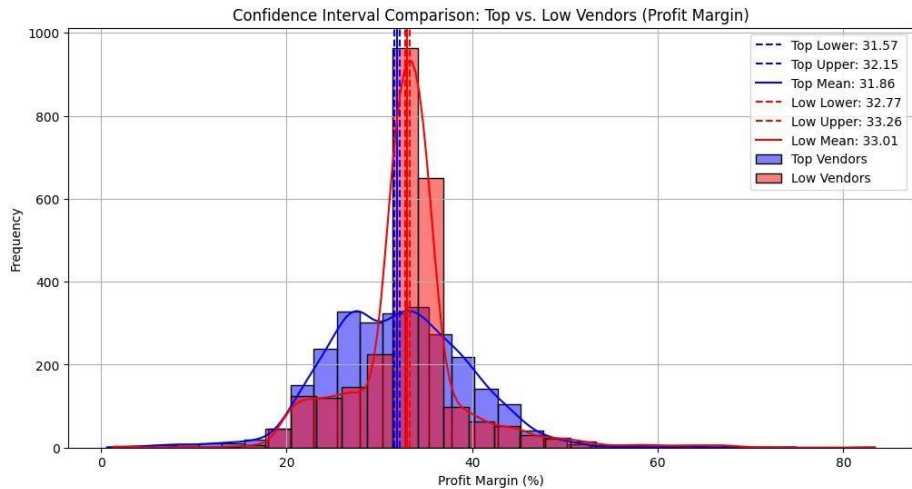


Figure 8. Confidence interval comparison: Top vs. Low vendors (profit margin).

Statistical Insight: Interestingly, the Low-Volume vendors actually maintain a higher average profit margin (~1.15% higher) than the high-volume giants. Since the confidence intervals do not overlap, this difference is statistically significant.

Business Implication: This confirms the "High-Volume, Low-Margin" nature of top players like DIAGEO. While they drive the bulk of the revenue, their margins are compressed due to competitive pricing and bulk trading. Conversely, the smaller vendors in the "Fragmented Tail" offer higher margins per unit, validating our previous strategy to target these "Hidden Gems" for growth.

6.2 Statistical Testing: Independent T-Test Analysis

To further solidify our confidence interval findings, we conducted a formal Independent Two-Sample T-Test to compare the mean profit margins of top-performing and low-performing vendors.

- **Null Hypothesis (H0):** There is no significant difference in the mean profit margins between the two groups.
- **Alternative Hypothesis (H1):** There is a statistically significant difference in their mean profit margins.

Metric	Value
T-Statistic	-5.9978
P-Value	0.0000 (highly significant)
Verdict	Reject H0

Since the p-value is far below the standard alpha threshold of 0.05, we reject the Null Hypothesis. This provides strong statistical evidence that the difference in profit margins between these two vendor groups is not due to random noise in the data.

Business Interpretation: The negative T-Statistic confirms the direction of the difference: Low-performing (smaller) vendors have a significantly higher mean profit margin than top-performing vendors. This validates the strategic pivot mentioned in Section 5.1 — while the Top 10 vendors provide the necessary volume for cash flow, the business must look toward smaller, niche vendors to drive higher profitability per unit.

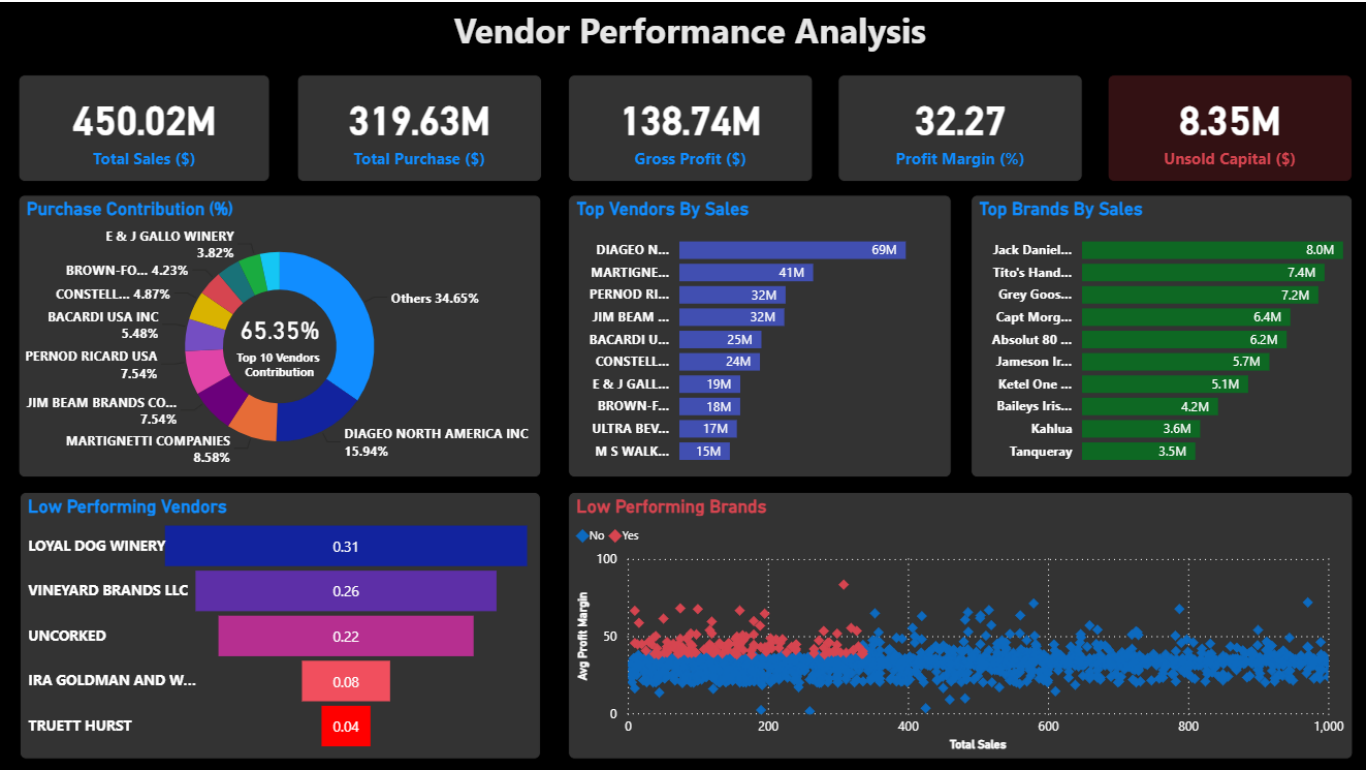
7. Vendor Performance Dashboard

The Vendor Performance Power BI Dashboard provides a consolidated view of financial performance, vendor contribution, and profitability. Key metrics show strong business performance with \$450.02M in total sales, \$138.74M gross profit, and a healthy 32.27% profit margin, while \$8.35M unsold capital highlights inventory optimization needs.

A Pareto analysis reveals that the Top 10 vendors contribute 65.35% of total procurement, indicating high dependency on a few key suppliers. Top vendors like DIAGEO and leading brands such as Jack Daniels drive the majority of revenue, ensuring stable cash flow. However, low-performing vendors exhibit poor inventory turnover, contributing to stagnant stock and tied-up capital.

The sales vs. profit margin analysis highlights a key trade-off: high-volume vendors generate consistent revenue with lower margins, while smaller vendors offer higher profitability per unit.

Figure 9. Vendor Performance Analysis dashboard summary.



Overall Insight: The dashboard emphasizes the need for a balanced procurement strategy — maintaining strong relationships with top vendors for volume while expanding high-margin, low-volume vendors to improve profitability and reduce risk.

8. Conclusion

The project demonstrates that while high-volume market leaders like DIAGEO provide the necessary cash flow, the true "margin drivers" are found within the fragmented tail of smaller vendors. The data-driven results lead to two primary strategic mandates:

- **Mandatory Order Consolidation:** To stop margin erosion, the procurement strategy must shift away from small, fragmented orders. Moving into the "Large" volume tier is the most direct lever for achieving the identified 73% cost savings.
- **Strategic Portfolio Balancing:** The business should protect its core volume with Top 10 vendors while aggressively expanding the "Hidden Gems" segment — brands with margins above 38% — to improve overall capital utilization.

By integrating advanced SQL aggregation, Python with statistics, this pipeline provides a robust framework for continuous inventory optimization and sustainable profit growth.